ECE101L lab1

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April 2026

1 Task 1: Verification of KCL and KVL

1.1 Introduction and Objective

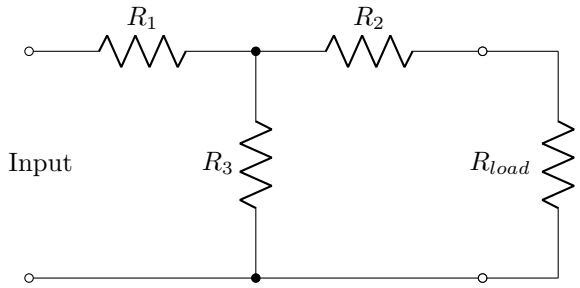


Fig 1

the circuit in Fig 1 is being used as the testing grounds for several circuit properties. Most notably, this is being used to prove Kirchhoff's current and voltage laws, abbreviated as KCL and KVL. After that, it will be used to compare the true resistors values to their nominal ones, and to compare the circuits calculated equivalent resistance to the real value. It will also be used to prove power consumed is equal to power supplied, and prove the current and voltage divider theorems. The use of an ohm meter is disallowed except for finding the circuits equivalent resistance.

1.2 Experiment

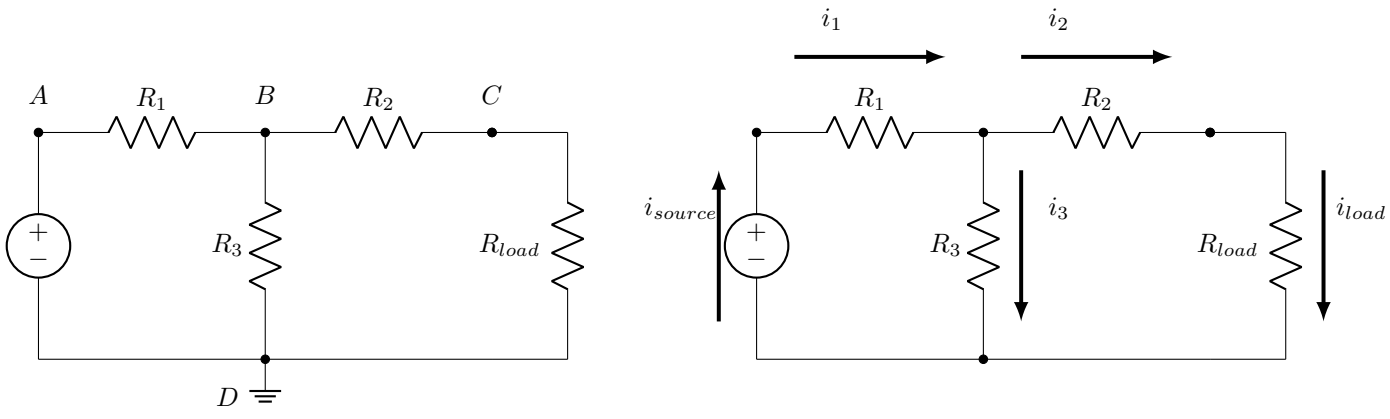


Fig 2

To begin, a $5V$ voltage source will be wired to the input terminals. Next all nodes on the circuit will be named, as well as naming all current paths. Fig 2 describes the new circuit and names. Ground has been

assigned to node D, this means it is our reference for zero volts. lastly all node voltages will be measured with a voltmeter, and each paths current with a galvanometer. These values are most of what is needed to prove the objectives.

Table 1

Path	Current
i_1	$33.642mA$
i_2	$16.827mA$
i_3	$16.830mA$
i_{load}	$16.827mA$
i_{source}	$33.642mA$

Node	Voltage
A	$4.999V$
B	$1.666V$
C	$3\mu V$
D	$0V$ (Ground)

Also note that Fig 2 has the correct current directions, and matches the measured directions.

1.2.1 KVL

Kirchhoff's Voltage Law states that the sum of voltage changes around any closed circuit loop must equal zero. This is because one node can not have two different voltages. When you complete a loop, the voltage should be the same value, therefore the sum of changes is zero.

To prove KVL, we will sum the voltage changes across components in a loop, $V_{after} - V_{before}$, and set the sum equal to zero. If all equalities are true, then it proves KVL. Three loop equations can be made on this circuit, seen in Fig 3. Starting the same loop from a different node will not change the resulting equation. And we should expect for this experiment that the sum of voltage change in a loop will equal approximately zero.

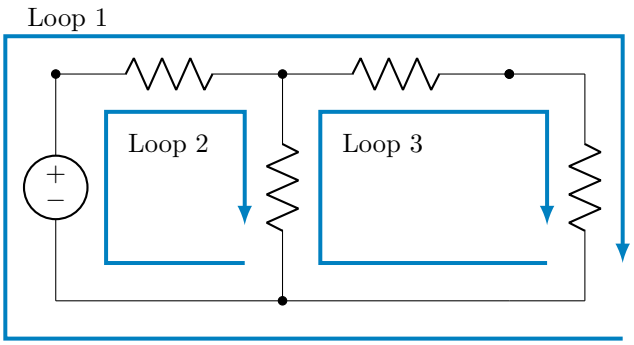


Fig 3

Loop 1: $(v_a - v_b) + (v_b - v_c) + (v_c - v_d) + (v_d - v_a) = 0$
Loop 2: $(v_a - v_b) + (v_b - v_d) + (v_d - v_a) = 0$
Loop 3: $(v_b - v_c) + (v_c - v_d) + (v_d - v_b) = 0$

Mathematically, these already prove the theorem since in every equation, each term repeats twice, once for positive and negative signage each. However if we simplify by solving each term in parentheses numerically it becomes even more clear.

Loop 1: $(3.333) + (1.666) + (3 * 10^{-6}) + (-4.999) = 0$
Loop 2: $(3.333) + (1.666) + (-4.999) = 0$
Loop 3: $(1.666) + (3 * 10^{-6}) + (-1.666) = 0$

The loop equations all approximate to zero, proving KVL. Loop 3 and 1 are off of zero by a tiny amount since I truncated the long decimal caused by subtracting $3 * 10^{-6}$, which is the voltage at C. Since the loops voltage changes are approximately zero, only off by small amounts, our prediction is true and KVL has been

verified.

1.2.2 KCL

Kirchhoff's Current Law states that the amount of current entering a circuit node is equal to the amount leaving. In other words, no current can be lost or generated at junctions. This is because electrical energy can't be created or destroyed. This is conservation of electrical energy.

To prove KCL, we will set the current entering a node equal to the current exiting for all four nodes. If all four equalities are true, then that proves KCL. Reference Fig 2 to see which currents are entering and exiting what nodes. We should predict that the current in is equal to current out.

Node A: $i_{source} = i_1$
Node B: $i_1 = i_2 + i_3$
Node C: $i_2 = i_{load}$
Node D: $i_{load} + i_3 = i_{source}$

This alone dose not prove the the theorem, however plugging in the numerical values shows that all statements are true.

Node A: $33.642mA = 33.642mA$
Node B: $33.642mA = 16.827mA + 16.830mA$
Node C: $16.827mA = 16.827mA$
Node D: $16.827mA + 16.830mA = 33.642mA$

The $15\mu A$ difference in node B and D from zero is small enough to excuse and can be explained by accuracy limits in the multimeter and electrical noise in the circuit. So these equations correctly prove KCL. Our predictions for the equations also were upheld.

1.2.3 Real and Nominal Resistance

Nominal resistance is the expected resistance of a resistor. But that value almost always varies from the real resistance. The Real resistance should be very close to the nominal value. The real resistance can be calculated with the formula $\frac{|v_a - v_b|}{i}$ where v_a and v_b are either of the adjacent nodes voltages, and i is the current passing through. The nominal voltage can be found by reading the color code on the resistors body, each of the colors refer to two digits, a multiplier, and a percentage tolerance. Fig 4 displays this color code. For this experiment we should expect all real resistances to be close to the nominal values and be within the tolerance.

R_1, R_2 and R_3 are all the same nominal resistance, with color code brown, black, brown, gold. This reads as 100Ω with a tolerance of $\pm 5\%$ tolerance. The tolerance indicates the real resistance should be between 95Ω and 105Ω. R_{load} is different since it is a 0Ω resistor, physically this translates to a short, or a wire, which has no color code.

Table 2

Resistor	v_a	v_b	i	Real	Nominal
R_1	$4.999V$	$1.666V$	$33.642mA$	99.072Ω	$100\Omega \pm 5\%$
R_2	$1.666V$	$3\mu V$	$16.827mA$	99.007Ω	$100\Omega \pm 5\%$
R_3	$1.666V$	$0V$	$16.830mA$	98.989Ω	$100\Omega \pm 5\%$
R_{load}	$3\mu V$	$0V$	$16.827mA$	$0.178m\Omega$	0Ω

Since the values are within the given tolerance range, we have verified that the resistors match the nominal values. The exception is R_{load} which had no tolerance, however the value is close enough to zero to

4 Band Resistors Color Codes Charts

Color	Color	1st Band	2nd Band	3rd Band Multiplier	4th Band Tolerance
Black		0	0	x1Ω	
Brown		1	1	x10Ω	±1%
Red		2	2	x100Ω	±2%
Orange		3	3	x1kΩ	
Yellow		4	4	x10kΩ	
Green		5	5	x100kΩ	±0.5%
Blue		6	6	x1MΩ	±0.25%
Violet		7	7	x10MΩ	±0.10%
Grey		8	8	x100MΩ	±0.05%
White		9	9	x1GΩ	
Gold				x0.1Ω	±5%
Silver				x0.01Ω	±10%

Fig 4

approximate it as such. This however is not proof that resistors are always in tolerance, but only that they can be expected to be inside it. When resistors get old or are over powered, they can become damaged and their resistance can change. This can become an issue if you accidentally pick a damaged resistor.

1.2.4 Conservation of Total Power

Power absorption is found with the equation $P = IV$, where power is in units of watts (W). Within circuits, power should be conserved, and so we can expect that the sum of power being absorbed is equal to the power supplied.

The passive convention tells us that if $P > 0$, then power is being absorbed, and if $P < 0$ power is being supplied. Since supplied power is the same as negative power absorption we can sum the power absorption of all components and set it equal to zero, if the equality is true then power is conserved.

To calculate the power, we will modify the equation to be $P = (v_a - v_b)i$. Where v_a and v_b are the adjacent node voltages to a component and i is the current traveling from v_a to v_b. Unlike the last section, v_a must be the more positive voltage. This is important since according to the passive convention, when power is being absorbed, current flows from high voltage to low. But if it is supplying power it moves from low to high.

For this experiment we may predict that the power being absorbed by the power source will be negative, I also expect power in R_1 to be greatest since it has the highest current available in the circuit, followed by R_2 tied with R_3. I think R_{load} will use the least power due to its very low voltage. And we can expect the sum of all their values to be zero.

Table 3

Component	v_a	v_b	i	P
R_1	$4.999V$	$1.666V$	$33.642mA$	$112.128mW$
R_2	$1.666V$	$3\mu V$	$16.827mA$	$28.033mW$
R_3	$1.666V$	$0V$	$16.830mA$	$28.038mW$
R_{load}	$3\mu V$	$0V$	$16.827mA$	$50.481nW$
V_{source}	$4.999V$	$0V$	$-33.642mA$	$-168.176mW$

Sum of Power: $0 = 112.128 + 28.033 + 28.038 + (50.481 * 10^{-6}) - 168.176$

The resulting Equality is off by $23.050\mu W$, however such a small difference is easy explainable by the small resistance in the wires, and or measurement error due to electrical noise. Since it's in the range of micro watts this explanation is plausible, and since it is such a small number its safe to approximate it as zero. This proves that the power is conserved. The expected power levels of all components of also was within prediction.

1.2.5 Equivalent resistances

An equivalent resistance is the value of resistance a network of resistors behaves as when wired together. If you replaced a large resistor network with a single resistor matching that value, nothing would change. That one resistor is called the equivalent resistor, often labeled R_{eq}. The resistance can be calculated or measured. Both these methods' results will be tried and compared. We can expect that both these methods will have close results.

To measure directly, simply remove the voltage source and connect an ohm meter to the input terminals. Doing so gives an equivalent resistance of 145.40Ω.

To calculate it we will use two formulas:

$$R_{series} = R_1 + ... + R_n$$
$$R_{parallel} = (\frac{1}{R_1} + ... + \frac{1}{R_n})^{-1}$$

Each is for calculating their respective configuration types. If a resistor is in parallel to a series combination or in series to a parallel combination that other combination must be reduced first. In our circuit we are only allowed to start with R_2 and R_{load}. The real resistances from Table 2 will be used for the calculation. The equation for our specific circuit is:

$$R_{eq} = R_1 + (\frac{1}{R_3} + \frac{1}{R_2 + R_{load}})^{-1}$$

The result of this is 148.57Ω. This value is 0.17Ω larger than the measurement using the ohm meter. This was very odd since I expected the value to be lower since it dose not account for wires. I was also expecting a much more marginal difference. I expect this to be due to accuracy and noise issues with my multimeter on small scales, similar to my the reason for error stated in the previous section. This also seems to disprove that the wires part of the issue in the last section.

Continuing, The difference seems acceptable. And the two methods did give similar results predicted.

1.2.6 Voltage and Current Dividers

When a voltage is applied across series resistors, the voltage is divided between the two resistors; the sum of the voltage across both resistors is equal to the voltage supplied. This is simply called the voltage divider theorem. Similarly, there is a current divider theorem that divides the applied current across parallel resistors. Both of these are fundamental current analysis rules. The equations for each are as follows:

$$V_n = \frac{R_n}{R_n + R_s} * V_{in}, \quad i_n = \frac{R_s}{R_n + R_s} * i_{in}$$

Fig 5 shows the simplest example of circuits that the theorems apply to. Either resistor can be picked as R_n in the equations. To test these equations, we need to re-represent our circuit as each of the circuits in Fig 5. We can expect that testing these equations will yield similar results to earlier measured values, and that our results will uphold to KVL and KCL as proven earlier.

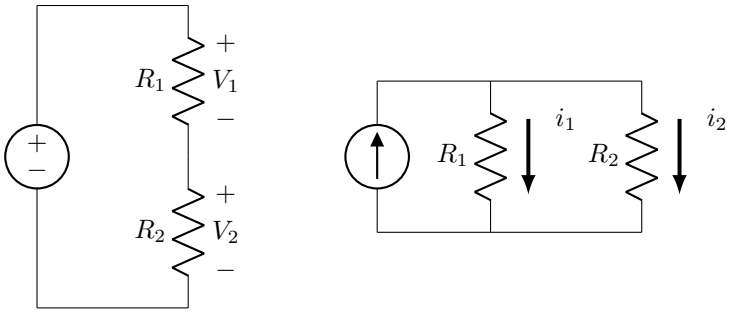


Fig 5

To find where current and voltage dividers apply to our circuit, we simply need to identify two series resistors and the voltage supplied to them, as well as two parallel resistors and the current supplied to them.

For the series, we can easily spot R_2 and R_{load}, but due to R_{load}'s low resistance, it makes it difficult to work with. Instead, we will use R_1 and the resistive equivalent of R_2, R_3, and R_{load}. Doing this will convert the circuit into the same configuration as the voltage divider circuit. The voltage source and R_1 are the same as the original testing circuit.

For the parallel resistors, we will use R_3 and the series combination of R_2 and R_{load}. To find the value of the current source, we will use the current entering the parallel area entering node b, and exiting node d. The value is the same as i_{source}. Fig 6 demonstrates our new circuit models.

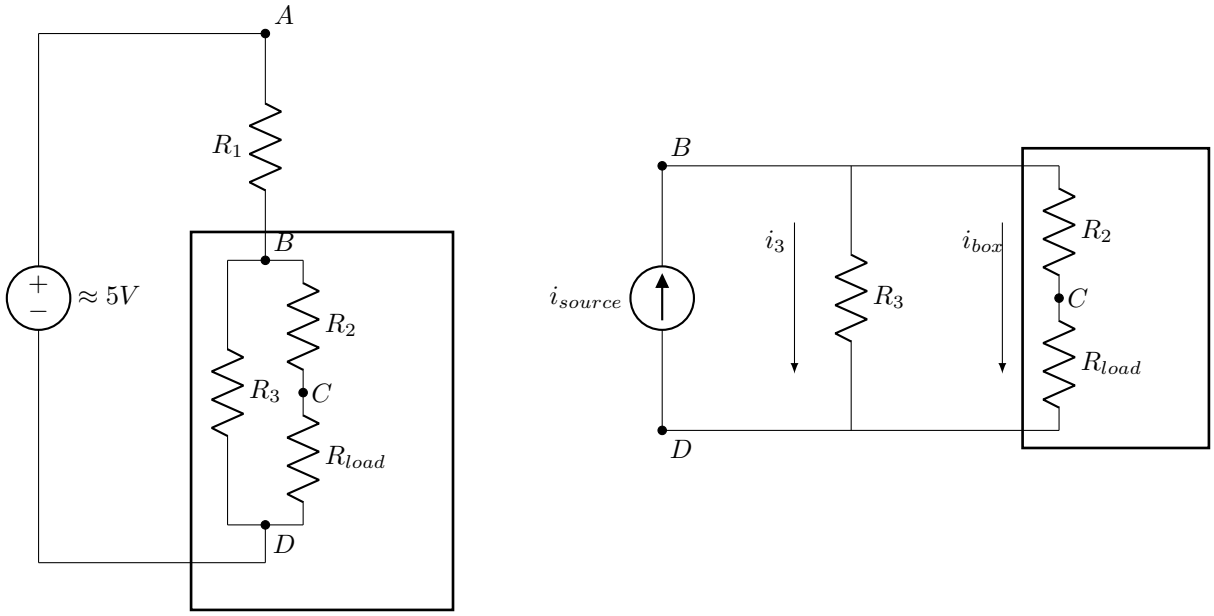


Fig 6

For the voltage divider, we can expect R_1 to have a larger voltage drop due to its higher resistance compared to the boxed area. The resistance of the boxed area R_{box} is 49.49Ω. Because of the earlier proof of KVL, we know that the sum of all voltage changes should be zero. Our equations for the voltage across both sections are:

$$v_1 = \frac{R_1}{R_1 + R_{box}} * 5V = 3.333V \qquad v_{box} = \frac{R_{box}}{R_1 + R_{box}} * 5V = 1.666V$$

These equations already match measured values.
The KVL equation becomes:

$$0 = 4.999V - v_1 - v_{box}$$

The KVL equation evaluates as true, numericaly its $0 = 4.99V - 3.333V - 1.666V$. Because the voltage divider equation demonstrates that it upholds circuit laws and that it matches values found in other ways, it proves the equation is true.

For the current dividers. We can expect that the currents i_{box} and i_3 are mostly equal due to the resistors being near equal as well. Here, $R_{box} \approx 99.007$. Using KCL as proved earlier, we can expect $i_{source} = i_3 + i_{box}$, and that i_{box} is the same as i_2. The equations are:

$$i_{box} = \frac{R_3}{R_{box} + R_3} * i_{source} = 16.819mA \qquad i_3 = \frac{R_{box}}{R_{box} + R_3} * i_{source} = 16.822mA$$

These are approximately the currents measured previously and are close to equal, as expected. This also makes the KCL equation true. Which numerically is $33.642mA = 16.822mA + 16.819mA$. Since the equation upholds previously proven circuit laws and was able to find previously measured values, it demonstrates that the current divider equation is true.

Both equations were able to return values accurate to what was found earlier. They also uphold to circuit rules found earlier, KCL and KVL. Because both equations uphold these earlier proven facts then it means they are true as well.

1.3 Conclusion

Using our circuit and measurements of it, we were able to prove multiple circuit laws and analysis tools including current and voltage dividers, KVL, and KCL. We were also able to prove that power is conserved, that equivalent resistance is real, and investigate the difference between nominal and real resistor values. All these things let us build a solid framework for understanding circuits and circuit analysis for future projects.

2 Task 2: Differential Voltages

2.1 Introduction and Objective

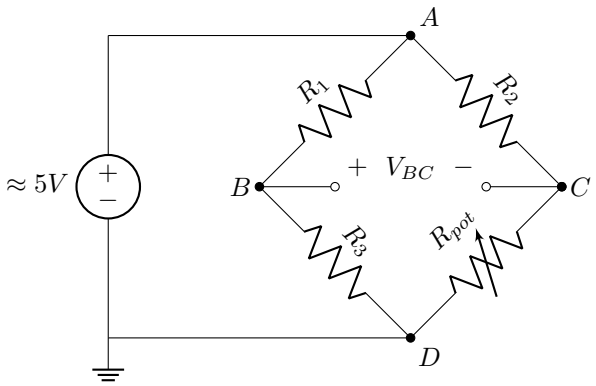


Fig 7

Differential Voltages are the differences between two single-ended voltages. Where single-ended voltages are made in reference to a common $0V$ node, Differential voltages have what is called a common mode voltage, labeled V_{cm}. The common mode voltage is the average of the single-ended voltages making up the differential. It is possible to split a single-ended voltage and create a differential situation. The circuit in Fig 7 will be used to demonstrate the relationship between single and differential voltages; it's called a Wheatstone Bridge. Inside, the differential voltage is directly measurable and is the voltage across nodes B and C, V_{BC}. To make sure we can get several various measurements, the fourth resistor is a potentiometer, or variable resistor, labeled R_{pot}. R_{pot} will let us alter the voltage on its branch since changing the resistance

will change the voltage.

For this section, there are three things we want to investigate.

- The resistances of R_1, R_2, and R_3 are measured by the DMM.
- The relationship between single-ended, differential, and common mode voltage.
- That the common mode voltage mainly depends on the single-ended voltage magnitudes, and that the differential voltage depends on the ratio of bridge resistances.

2.2 Experiment

2.2.1 Measured Resistances

This task is straightforward. Just comparing the resistances as measured by the ohmmeter to the nominal resistances. I will also include the variable resistors' nominal value.

Table 4

	R_1	R_2	R_3	R_{pot}
Nominal	$560\Omega \pm 5\%$	$560\Omega \pm 5\%$	$560\Omega \pm 5\%$	0Ω to $1k\Omega$
Measured	558.5Ω	557.2Ω	559.9Ω	0.2Ω to $1.082k\Omega$

Just like the resistance experiment in the first task, all the resistances fit within their tolerance, which is good. The exception is R_{pot} with no given tolerance. This data doesn't prove anything but is important to these later parts.

2.2.2 Differential and Single-ended Voltage Relationships

The relationship between Single-ended, differential, and common mode voltages should all be explainable using our circuit. These relationships are given by $V_{cm} = \frac{V_1 + V_2}{2}$ and $V_{diff} = V_1 - V_2$. We will be verifying these relationships well as finding any rules in the relationship between these values. I think that V_{cm} will not be verifiable, since it can not be directly measured on the circuit. V_{diff} will be verifiable with the earlier equation and will get similar values to what we can measure. I also think we will be able to find a clean set of rules to describe the relationships.

To start, several measurements have been made for all our voltages and the resistance of R_{pot} for several trials.

Table 5

V_B	V_C	V_{diff} (Measured)	R_{pot}
$2.495V$	$1.060mV$	$2.494V$	0.2Ω
$2.495V$	$0.698V$	$1.797V$	90.49Ω
$2.496V$	$1.393V$	$1.101V$	213.64Ω
$2.495V$	$1.966V$	$0.530V$	363.10Ω
$2.495V$	$2.501V$	$-5.600mV$	561.25Ω
$2.496V$	$2.884V$	$-0.386V$	762.75Ω
$2.496V$	$2.992V$	$-0.495V$	836.49Ω
$2.496V$	$3.294V$	$-0.798V$	$1.082k\Omega$

Now, using this data, we can use the formulas for common mode voltage $V_{cm} = \frac{V_b + V_c}{2}$ to calculate the common mode voltage for each trial. The differential voltage will also be calculated, which is simply the difference between voltages $V_{diff} = V_B - V_C$. We can use this to verify the measured values.

Table 6

V_{cm}	V_{diff}
1.248	$2.494V$
1.597	$1.797V$
1.945	$1.103V$
2.231	$0.539V$
2.498	$-6.0mV$
2.690	$-3.88V$
2.744	$-0.496V$
2.895	$-0.798V$

The calculated values for V_{diff} are all only one or two millivolts away from the measured values, indicating the measurements and calculations are likely correct.

The key relationship that the common mode voltage has to the single-ended ones is that they describe the average of the two nodes from ground. This is easy to spot from the earlier formula. Inside components like amplifiers, if the common gets too high, then it can lead to distortion.

The differential is simple since it's just the difference of the voltages. Also, notice that if it becomes negative, it means the second node's voltage is overtaking the voltage of the first. otherwise its positive.

It can also be observed that as the common voltage rises, the differential drops. When the single-ended voltages approach the same value, the common mode approaches that value. However, this may not have been the case if both branches were changing.

These relationships are easy to spot from trial 6, where $V_{diff} = -6.0mV$ and $V_{cm} = 2.498V$. The voltages of the two nodes were very close to 2.498, which made the differential approach its closest to $0V$. After, as node C rose higher than B, the difference became negative while the common voltage continued to rise. Below trial 6, the average became lower, and eventually the differential value became approximately the same as node B

A few simple rules for any differential situation can be taken from all these observations, where V_1 is the general name of our V_B and V_2 is the general name of our V_C:

- As V_1 and V_2 approach the same value, V_{diff} approaches $0V$
- As V_1 and V_2 approach the same value, V_{cm} approaches the same value
- One of two inequalities is always true, $V_1 > V_{cm} > V_2$ or $V_2 > V_{cm} > V_1$
- When the first inequality is true, V_{diff} is positive, if the second is true, V_{diff} is negative.
- V_{diff} can not exceed V_1 or fall below $-V_2$.
- Since V_{diff} and V_{cm} are in terms of V_1 and V_2, we can write V_1 and V_2 in terms of V_{diff} and V_{cm}. $V_1 = V_{cm} + \frac{V_{diff}}{2}$ and $V_2 = V_{cm} - \frac{V_{diff}}{2}$.

Since we were able to describe a set of rules for the relationships of the three voltage types, it indicates that the relationships of the three are strongly correlated. The ability to find matching numbers for the measured and calculated differential voltage also validates its equation and relationship to single-ended voltages.

2.2.3 Relationships to Ratios and Magnitudes

The differential voltage and common mode voltage are assumed to be related to specific values of the circuit. The differential is thought to be relatable to the ratios of resistances in the bridges. The common mode is thought to be relatable to the magnitudes of single-ended voltages.

I believe that observing changes these values are making over time will let us verify these assumptions. And I believe that both types of voltages will match these assumptions.

To show how the differential voltage is related to the bridge resistor ratios, we will use the voltage divider equations for V_1 and V_2. Since the two halves of the bridge are series resistors it will allow us to relate the voltages and resistances easily.

$$V_1 = \frac{R_3}{R_3 + R_1} * V_{in} \quad \text{and} \quad V_2 = \frac{R_{pot}}{R_{pot} + R_2} * V_{in}.$$

entering these into the equation for V_{diff} gives:

$$V_{diff} = \left(\frac{R_3}{R_3 + R_1} * V_{in}\right) - \left(\frac{R_{pot}}{R_{pot} + R_2} * V_{in}\right)$$

Now simplifying slightly gives:

$$V_{diff} = \left(\frac{R_3}{R_3 + R_1} - \frac{R_{pot}}{R_{pot} + R_2}\right) * V_{in}$$

Voltage dividers relate the ratios of resistors to the total resistance present with voltage. Since the bridge is just two of these voltage dividers, we can easily show that the differential voltage is closely related to the resistance ratios. The differential voltage is a display of how imbalanced the resistances in each branch become. If they diverge, the differential deviates from zero, but the differential will converge to zero the more similar the branch ratios are.

You can also find what sign the differential will be by using the ratio $\frac{V_1}{V_2}$. If it's one or close to it, the differential will be approximately zero. If the ratio is less than one, the differential is negative. And if the ratio is larger than one, the voltage is positive.

It is also possible to express the common-mode voltage using voltage dividers, but since the common mode is an average, its value is not sensitive to small changes in the ratio. You can observe from the test data that it changed much more slowly than the differential voltage. The differential voltages' total change was $3.292V$, the common voltages' total change was $1.647V$. The main reason it can't be explained like earlier is that the common voltage provides no information about the ratio mismatch. Instead, we should look at what an average actually is. It is the exact middle between a set of values, two in our case. And since averages are magnitude reliant that means the common mode voltage is as well

By knowing the middle value, we can understand how large our voltages are. Recall the previous inequalities, $V_1 > V_{cm} > V_2$ or $V_2 > V_{cm} > V_1$. These are due to the nature of averages, and because of it, we always know where the two voltages are relative to the common. With the properties of DC circuits, we know that the single-ended voltages can't exceed the source voltage in a DC circuit, and the voltages should not be negative relative to a properly placed ground node. Using this, we can modify the inequality to become, $V_{source} > V_{1or2} > V_{cm} > V_{2or1} > 0V$.

This inequality also demonstrates with more proof how common mode is not ratio based, since V_1 and V_2 being larger than one another has no effect on the final value, unlike the differential voltage.

lastly lets rewrite the equations $V_1 = V_{cm} + \frac{V_{diff}}{2}$ and $V_2 = V_{cm} - \frac{V_{diff}}{2}$, as $V_{cm} = V_1 - \frac{V_{diff}}{2}$ and $V_{cm} = V_2 + \frac{V_{diff}}{2}$. Recall that the single ended voltage can't drop below zero, and note that the differential can't surpass V_{source} or drop under $-V_{source}$. Each input voltage can be viewed as consisting of a shared component plus a small deviation. By expressing V_1 and V_2 as a common-mode term plus and minus half the differential voltage, the common-mode voltage is isolated as the quantity that represents the overall magnitude common to both signals.

Using a rewritten form of the differential voltage allowed us to confirm its relationship to bridge resistors ratios as well as to better describe what that relationship is. Similarly, for the common mode voltage, using the properties of averages and observations of its behavior, we were able to better describe its relationship to single-ended voltage magnitudes.

2.3 Conclusion

Using several trials of measurements, we were able to track how three ways of measuring voltage, common mode, differential, and single-ended, and describe their relationships to each other in a circuit. Investigating this relationship let us find a simple set of rules that they all follow. We were also able to investigate the common mode and differentials voltage to parts of the circuit and found the differential relies heavily on the ratio between the resistances of the bridges, and that the common mode relies on the magnatudes of the single-ended voltages.